

7 Read the passage below and answer the questions that follow.

Modern aircrafts operated by airlines use jet engines. A jet engine is a type of reaction engine, drawing in air and discharging a fast-moving jet of heated exhaust gas (air mixed with jet fuel) that generates thrust by jet propulsion.

Fig. 7.1 shows an engine designed for modern passenger aircraft that draws in a maximum mass of air of 1300 kg per second. The diameter of the engine intake fan is 3.0 m. The air drawn into the engine is compressed, or “squeezed”, and then in the combustion chamber, it is heated to a very high temperature. Finally, the engine blows the heated exhaust gas back out of its nozzle producing thrust.

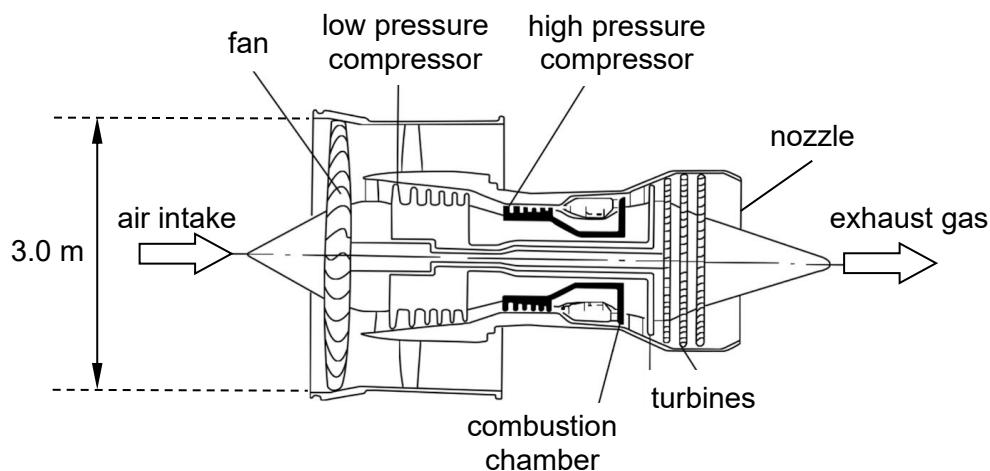


Fig. 7.1

The thrust F produced by such an engine is given by the expression

$$F = (m_{\text{air}} + m_{\text{fuel}})v_{\text{exhaust}} - m_{\text{air}}v_{\text{in}}$$

where m_{air} is the mass of air intake per second, m_{fuel} is the mass of fuel consumed per second, v_{in} is the velocity of air intake, and v_{exhaust} is the velocity of exhaust gas.

Table 7.1 show data for the engine operating during take-off.

Table 7.1

m_{air}	1300 kg s ⁻¹
m_{fuel}	17 kg s ⁻¹
v_{exhaust}	410 m s ⁻¹

A modern passenger aircraft is fitted with two such engines. When operating at maximum thrust, the aircraft with the maximum take-off weight (MTOW) of 2.75×10^6 N is accelerated from rest to an airspeed that provides sufficient lift to be airborne in 45 s.

At take off, the aircraft performs a rotation to point its nose upwards for take off as shown in Fig. 7.2. This is accomplished with the aid of its rear horizontal stabilisers as shown in Fig. 7.3.

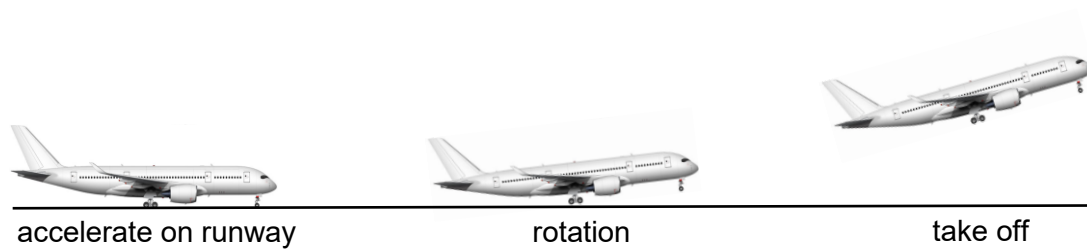


Fig. 7.2

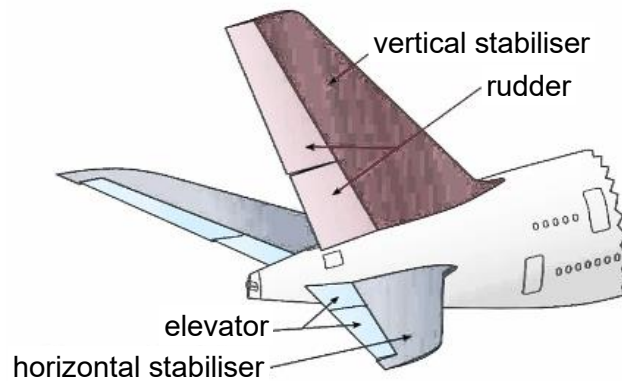


Fig. 7.3

The energy density of jet fuel is 42 MJ kg^{-1} .

Aviation accounts for about 2.5% of global carbon dioxide emissions, so transitioning the sector away from its dependence on fossil fuels to clean electric power is seen as a priority. While battery-powered electric aircraft is far from being able to replace the large passenger aircraft used on long-haul routes, they may be able to take on city-to-city hops of several hundred kilometres.

The electrical energy onboard aircrafts is stored in lithium-ion batteries. When fully charged, the battery has a specific energy of $0.260 \text{ kW h kg}^{-1}$.

(a) (i) The density of air is 1.2 kg m^{-3} .

Show that the velocity of air intake of the engine is 150 m s^{-1} .

[2]

(ii) Calculate the maximum thrust F .

$$F = \dots\dots\dots \text{ kN [2]}$$

(iii) Determine the maximum acceleration of the aircraft.

$$\text{maximum acceleration} = \dots\dots\dots \text{ m s}^{-2} [2]$$

(b) A cross-section of the rear horizontal stabiliser on the aircraft during rotation for take off is shown in Fig. 7.4.

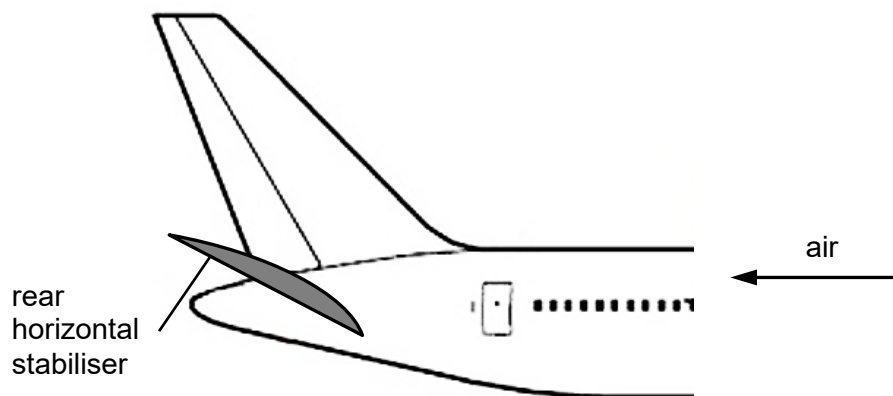


Fig. 7.4

The stabiliser deflects air when the aircraft is moving along the runway.

Explain, by reference to the change in momentum of the air, why this air flow causes the aircraft to rotate for take off.

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..... [2]

(c) Some of the forces acting on the aircraft while accelerating from rest are shown in Fig. 7.5.

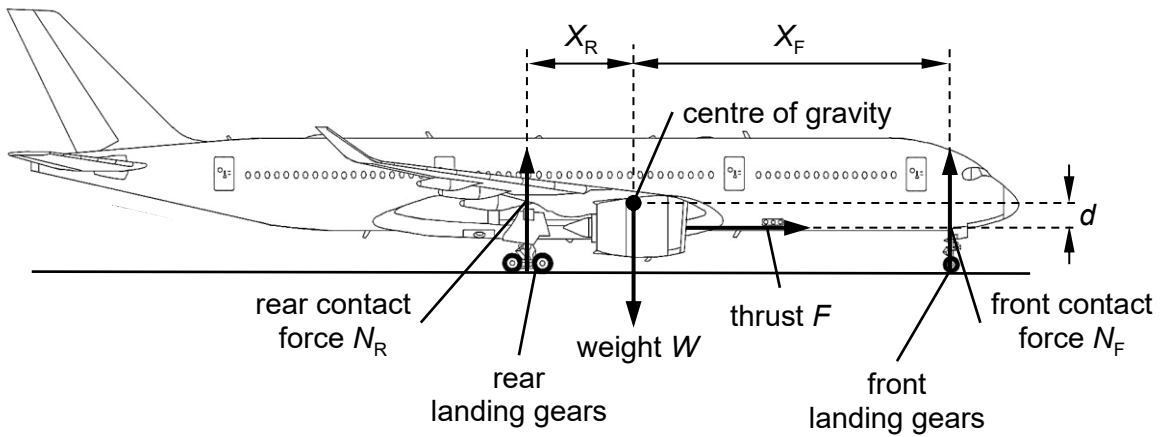


Fig. 7.5

N_F and N_R are the contact force acting on the rear and front landing gears respectively. W is the weight of the aircraft. F is the thrust acting on the aircraft by the exhaust gas.

The centre of gravity is a vertical distance d above the line of action of F . The centre of gravity is a horizontal distance X_R from the rear landing gears and a horizontal distance X_F from the front landing gears.

For the aircraft accelerating from rest:

(i) take moments about the centre of gravity, to show that

$$N_R = \frac{WX_F + Fd}{X_F + X_R}$$

[3]

(ii) determine a similar expression to that in (c)(i) for N_F .

[1]

(iii) Use your expressions in (c)(i) and (c)(ii) to show that there is weight transfer between the front and rear landing gears when the aircraft is accelerating from rest.

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 [2]

(d) A passenger aircraft consumes jet fuel at a rate of $5.5 \times 10^3 \text{ kg h}^{-1}$ when cruising. A flight from Singapore to Munich takes 12 hours.

(i) Calculate the total energy used during cruising flight of 12 hours.

total energy = J [2]

(ii) Suggest what is meant by a specific energy of $0.260 \text{ kW h kg}^{-1}$.

.....

 [1]

- (iii) Determine the mass of the lithium battery needed to supply the total energy in (d)(i).

mass = kg [2]

- (e) (i) Use your answer in (d)(iii) to explain if the aircraft can be purely battery-powered for a flight from Singapore to Munich.

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 [1]

- (ii) Suggest two advantages that the jet engine aircraft has over the battery-powered electric aircraft.

1.

 2.
 [2]

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